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Set 13: Geographic Data

Question 1: Map Chart

Aim

To create a map chart showing the distribution of cities and label its elements.

Procedure

1. Load ggplot2 and create the data frame (with assumed longitude and latitude).
2. Plot cities as points on a longitude-latitude map, sized by population.
3. Add city labels, title and axis labels, then display the map.

Code

```
library(ggplot2); library(gridExtra)
d <- data.frame(City=c("City A","City B","City C"),
  Population=c(500000,700000,600000), Temp=c(75,68,80),
  Elevation=c(1000,800,1200), Lon=c(77,80,73), Lat=c(13,19,26)) # lon/lat
assumed
m <- ggplot(d, aes(Lon, Lat, size=Population, label=City)) +
  geom_point(color="red") + geom_text(vjust=-1.5, size=3) +
  labs(title="Distribution of Cities", x="Longitude", y="Latitude")
print(m)
```

Output

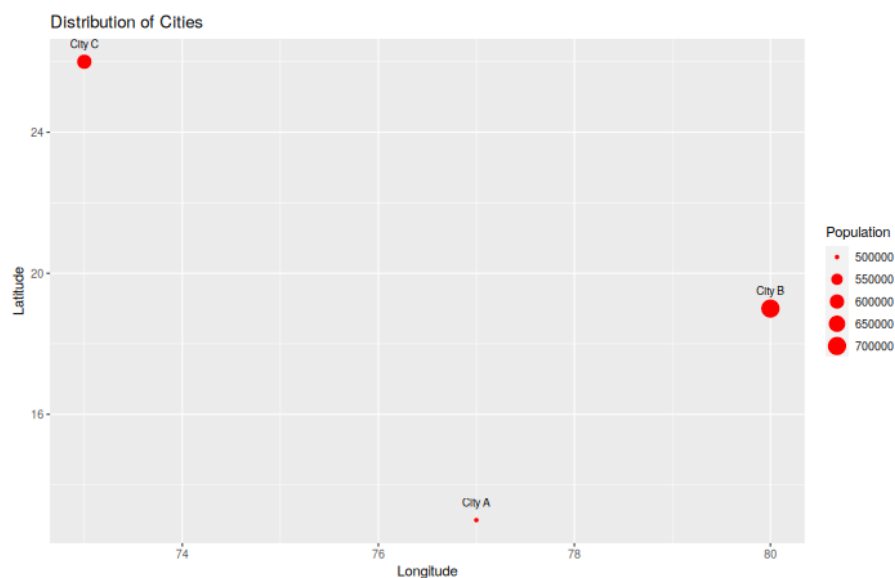


Fig 1: Map chart of cities

Result

The map chart was created. Each city is a labelled point; larger points mean larger population (City B is the biggest).

Question 2: Scatter Plot

Aim

To explore the relationship between average temperature and population size using a scatter plot.

Procedure

1. Plot Temperature on the x-axis and Population on the y-axis.
2. Add city labels and a trend line.
3. Find the correlation with `cor()` and interpret it.

Code

```
s <- ggplot(d, aes(Temp, Population)) + geom_point(size=4, color="blue") +  
  geom_smooth(method="lm", se=FALSE) + geom_text(aes(label=City), vjust=-1)  
+  
  labs(title="Temperature vs Population", x="Avg. Temperature (F)",  
y="Population")  
print(s)  
cat("Correlation:", round(cor(d$Temp, d$Population), 2))
```

Output

Console output:

Correlation: -0.58

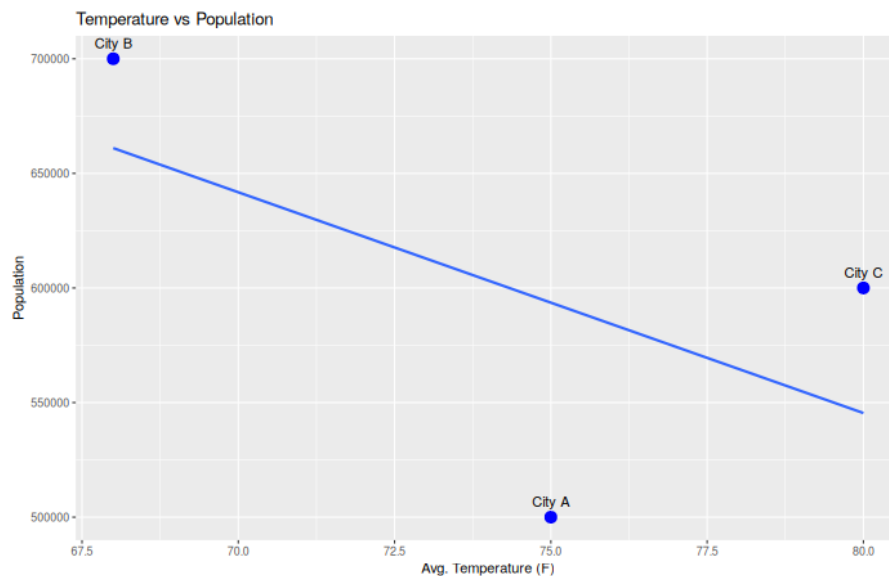


Fig 2: Scatter plot – Temperature vs Population

Result

The scatter plot was created. Insight: the correlation is -0.58 (negative), so the hotter cities (C: 80°F, A: 75°F) have smaller populations than the coolest city B (68°F, 700000). With only three cities this is a weak indication.

Question 3: Table

Aim

To build a labelled table displaying the geographic data of each city.

Procedure

1. Select the City, Population, Temp and Elevation columns.
2. Print the data frame and draw it as a table using tableGrob().

Code

```
print(d[, 1:4])  
t <- tableGrob(d[, 1:4], rows=NULL)  
grid.arrange(t)
```

Output

Console output:

```
City Population Temp Elevation  
1 City A      500000 75      1000  
2 City B      700000 68        800  
3 City C      600000 80      1200
```

City	Population	Temp	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Fig 3: Table of geographic data

Result

The table displaying city, population, average temperature and elevation was created with labelled column headings.

Question 4: Dashboard

Aim

To combine the map chart, scatter plot and table into one dashboard for exploring the geographic data.

Procedure

1. Use the map (m), scatter plot (s) and table (t) from the earlier questions.
2. Arrange all three in one panel using `grid.arrange()`.

Code

```
grid.arrange(m, s, t, ncol=2)
```

Output

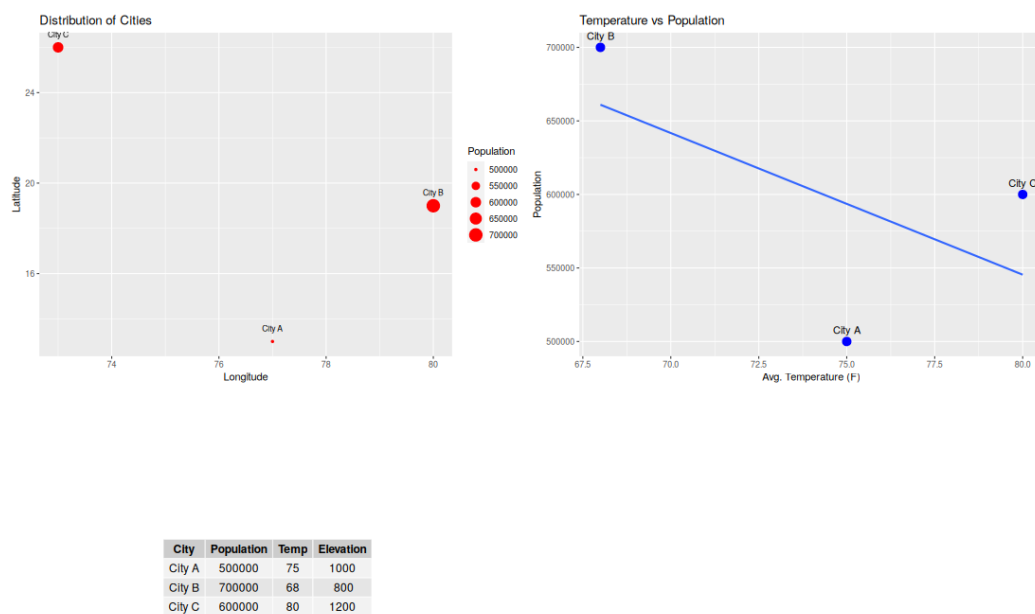


Fig 4: Dashboard (map + scatter plot + table)

Result

The dashboard combining the map chart, scatter plot and table was created in R.